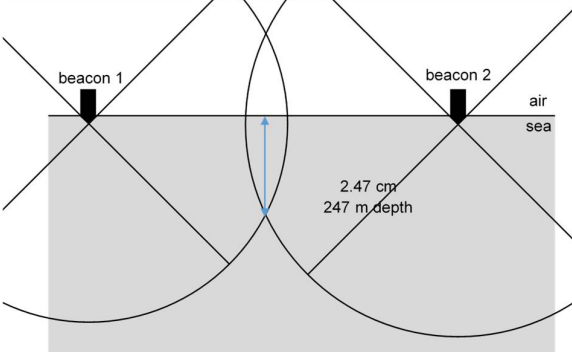


	r_p	
7(a)(i)	An object will <u>continue</u> to be in a state of rest or move with <u>constant velocity</u> , unless acted upon by a <u>net external force</u> .	B1
7(a)(ii)	[sensor output] when <u>acceleration</u> measured is 0, the vehicle can be either <u>stationary</u> or moving with <u>constant velocity</u> (inferred from Newton's first law) [effect on position] the position can be <u>fixed</u> or <u>constantly changing</u> OR can <u>only</u> infer whether there is a <u>change</u> in position	B1 B1 (B1)
7(b)(i)	[magnitude] rate of change of momentum of a body is directly proportional to resultant force acting on it [direction] takes place in the direction of the resultant force	B1
7(b)(ii)	let proof mass be of mass m each spring constant to be k acceleration of vehicle be a $ma = 2kx$ $x = \left(\frac{m}{2k} \right) a$	B1 (2k) B1 ($x \propto a$)

7(b)(iii)1.	no oscillations take place, displacement brought to zero in shortest possible time	B1
	oscillation will result in random errors in the form of positive and accelerations when moving vehicle no longer accelerates	B1
7(b)(iii)2.	frequency at which a system vibrates at in the absence of resultant external forces	B1
	avoid resonance from sources of external periodic driving forces such as engines	B1
7(b)(iii)3.	for the same acceleration, higher natural frequency means less displacement x , hence less sensitive	B1
	$a = -\omega^2 x = -(2\pi f)^2 x$ OR higher natural frequency means the spring has a higher spring constant and hence x is lower with the same amount of force applied	(B1)
7(c)(i)	compresses a large range of a <u>few orders of magnitude</u> to be shown on the same graph	B1
7(c)(ii)1.	first data point is time interval for single data point $f = \frac{1}{T} = \frac{1}{0.1} = 10 \text{ Hz}$	A1
7(c)(ii)2.	0.0035 m s ⁻² (to half a division accuracy)	B1
7(c)(iii)	let Allan Deviation be σ show $\sigma = \frac{k}{\sqrt{t}}$ i.e. $\sigma\sqrt{t} = k$	M1
	consider (2, 0.001), $k_1 = 0.001\sqrt{2} = 0.0014142$ consider (5, 0.0007), $k_2 = 0.0007\sqrt{5} = 0.0015625$ consider (7, 0.0006), $k_3 = 0.0006\sqrt{7} = 0.0015875$ largest percentage deviation = $\frac{k_3 - k_1}{k_1} \times 100\% = 12\%$ compare deviation against a reasonable percentage specified by candidate (must be below 20%) (minimally: to pick 2 points to compare)	A1
7(c)(iv)1.	a single time duration between 100-200 s	B1
7(c)(iv)2.	0.0003 m s ⁻²	B1

7(c)(iv)3.	$s = ut + \frac{1}{2}at^2$ $\Delta s = 0 + \frac{1}{2}(0.0003)(3600^2)$ $= 1900 \text{ m}$	C1 A1
7(d)(i)	 (No mark if concentric circular loci cannot be seen.) 200 – 250 m	B1 (concentric circular loci) B1
7(d)(ii)	disrupt marine animals which rely on sonar to hunt and navigate	B1